

Media-in-the-loop holography: designing photopolymer exposures through a differentiable recording model

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Abstract: Camera-in-the-loop correction cannot be applied to write-once volume photopolymers, whose reconstruction error is measured only once it is permanent. We study, in simulation, optimizing the recording exposure through a differentiable non-local photopolymerization-diffusion model with beam-propagation readout. On an unslanted grating benchmark of 30 spatial-frequency and contrast-budget cells, the gain over a media-blind optimizer is positive in 29 cells (at most 2.52 dB) and decays toward zero at high spatial frequency; the share of cells more than 2 dB below an oracle reference falls from 37% to 0%. A slope-matched linear-recording control shows that the gain requires a nonlinear dose response, and it largely vanishes for fringes slanted by 10° or more.

1. Introduction

Phase-only computer-generated holography (CGH) computes a modulation pattern whose diffracted field approximates a target. When the modulator departs from its model, camera-in-the-loop (CITL) optimization measures the reconstruction and back-propagates the residual into the display model [1]; related approaches refine the model with Wirtinger gradients [2], learned proxies [3] or time-multiplexed control [4]. All of them rely on a rewritable modulator.

Volume photopolymers such as Bayfol HX [5], acrylamide/PVA and PQ/PMMA [6] record a permanent index distribution through photopolymerization and mass transport, and the recorded index is not a scaled copy of the exposure: monomer diffusion [7,8], non-local chain growth [9,10], dye bleaching and a saturating index response act nonlinearly [11]. Because the medium is written once, a CITL-style measurement arrives after the error is fixed, and the recording model must carry the correction without feedback.

Differentiable models of a fabrication process already sit inside optical design loops: neural lithography learns a photolithography simulator for diffractive optics [12], and adjoint inverse design treats a physical forward model as a differentiable layer [13]. Concurrently, Wechsler et al. [14] coupled a differentiable wave-optical model to a photochemical model for holographic volumetric printing and demonstrated printed parts, an experiment this study does not attempt; we make no priority claim for differentiable recording models in a design loop.

This paper asks, in simulation, how much an exposure optimized through a differentiable model of photopolymer recording (“media-in-the-loop”, MIL) improves on the same optimizer run with a linear, media-blind recording assumption, for a volume index grating scored by its reconstruction, and where the improvement comes from. The recording model (non-local polymerization-driven diffusion, NPDD), the readout (split-step beam propagation) and the optimizer are established; the contribution is their combination into an exposure-design loop and its characterization across spatial frequency, contrast budget, recording mechanism, grating slant and twin miscalibration. Every headline number is a paired difference (MIL minus media-blind design) with both arms designed and scored on the same simulated medium, the “twin”. An exposure-independent calibration error of the twin cancels in that difference; an exposure-dependent one does not, so we also design on the nominal twin and record on miscalibrated ones (Section 4.7) and compare

the twin with published data (Section 5). All results are simulation outputs of this study unless cited, and every number is generated by script from released per-job result files. This is the second part of a two-part study; Part 1 characterized phase-only CGH degradation with idealized modulators [15].

2. Model

Table 1 lists every constant; the released code is the reference implementation.

2.1. Exposure and feasible set

The design variable is the delivered exposure $E(x) \geq 0$ on the transverse coordinate x , constrained to a mean dose E_0 and a peak-to-mean contrast budget B_c :

$$\mathcal{E}(E_0, B_c) = \left\{ E : E \geq 0, \bar{E} = E_0, \max_x E(x)/\bar{E} \leq B_c \right\}. \quad (1)$$

After every optimizer step the exposure is projected onto $\mathcal{E}$ by 20 alternating passes of dose rescaling, $E \leftarrow E E_0/(\bar{E} + \epsilon)$, and peak clipping, $E \leftarrow \min(E, B_c E_0)$. Both operations are elementwise and differentiable; the residual relative constraint violation after 20 passes is below 10^{-6} on the tested targets.

2.2. Recording: non-local polymerization-driven diffusion

Free monomer $u(x, t)$ ($u(x, 0) = 1$), polymer $N(x, t)$ ($N(x, 0) = 0$) and unbleached dye $d(x, t)$ ($d(x, 0) = 1$) evolve as in the NPDD model [9], itself built on the local model of Zhao and Mouroulis [7]:

$$\partial_t u = \partial_x (D_{\text{eff}}(N) \partial_x u) - G_\sigma * (Fu), \quad (2)$$

$$\partial_t N = G_\sigma * (Fu), \quad (3)$$

$$\partial_t d = -k_{\text{bleach}} I d, \quad (4)$$

with $F = \kappa (Id)^\gamma$, recording intensity $I \propto E$, a Gaussian non-local kernel G_σ of width σ ($\widehat{G_\sigma}(K) = e^{-K^2 \sigma^2/2}$) and $D_{\text{eff}}(N) = D_0 e^{-\alpha_D N}$. The kernel acts on the production term Fu , so that polymer initiated at x' is deposited at x [9]. The recorded index modulation is

$$\Delta n(x) = \Delta n_{\text{max}} \tanh(\beta N(x)), \quad \beta = 1.5. \quad (5)$$

The saturating map and the value of β are modelling choices of this work, held fixed in every fit. Three mechanisms saturate the dose response: monomer depletion ($u \rightarrow 0$), dye bleaching ($d \rightarrow 0$) and the tanh map; Section 4.5 separates them.

2.3. Integration and gradients

Equations (2)–(4) are integrated over $t_{\text{total}} = 10$ s in $n_{\text{steps}} = 300$ first-order implicit–explicit steps. The explicit reaction step evaluates the non-local production by fast Fourier transform, caps consumed monomer at the monomer available (so $u \geq 0$ and $u + N$ is conserved to rounding error) and updates d exactly for fixed I . The implicit diffusion step applies $\exp(-\bar{D} K^2 dt)$ in Fourier space with $\bar{D}$ the spatial mean of D_{eff} ; this approximation to Eq. (2) has a one-step error of about 0.1% of the scale of u at a representative late state, and its accumulation over a trajectory was not bounded. Gradients are the discrete adjoint of this integrator, obtained by reverse-mode automatic differentiation [16, 17].

2.4. Readout: split-step beam propagation

The recorded profile $\Delta n(x)$ is extruded uniformly through a slab of thickness T and read out at wavelength λ by scalar split-step beam propagation [18]: $n_z = 128$ slices, each applying the phase $\exp[ik_0 \Delta n dz(1-s)]$ ($k_0 = 2\pi/\lambda$, shrinkage s) followed by band-limited angular-spectrum propagation [19], then one angular-spectrum step to the reconstruction plane, where the intensity is the reconstruction R . The periodic window represents an infinite grating. Slanted fringes ($x - z \tan \phi = \text{const}$) are represented by a per-slice lateral Fourier shift. All headline results use an unslanted grating ($\phi = 0$), the geometry in which scalar theory is closest to rigorous coupled-wave analysis (RCWA) [20] (Section 6); slant is studied separately (Section 4.6). Between $n_z = 128$ and 256 the paired gain changes from 0.915 to 0.894 dB at $K = 1.96 \text{ rad}/\mu\text{m}$ and from 0.900 to 0.928 dB at $K = 3.93 \text{ rad}/\mu\text{m}$ (one seed, Supplement 1, Section S3). The Klein–Cook parameter $Q = 2\pi\lambda T/(n_0\Lambda^2)$, $\Lambda = 2\pi/K$ [21], crosses 10 at $K \approx 2.79 \text{ rad}/\mu\text{m}$, so the two lowest tested K lie in the intermediate regime. Kogelnik theory [22] is used only as a closed-form cross-check.

2.5. Loss and optimizer

With target intensity I_t on n_x samples, the loss is the scale-invariant mean-squared error

$$\mathcal{L}(R, I_t) = \frac{1}{n_x} \|\alpha^\star R - I_t\|_2^2 / \max(I_t)^2, \quad \alpha^\star = \langle R, I_t \rangle / \langle R, R \rangle, \quad (6)$$

and we report $\text{PSNR}_{\text{si}} = -10 \log_{10} \mathcal{L}$, a monotone transform of the optimized objective. The optimizer is Adam [23] (learning rate 5×10^{-2} , 800 iterations, convergence tolerance 10^{-4}) on the parameterization $E = \text{softplus}(\theta) + 10^{-6}$. The random seed enters only through the initialization $\theta_0 = 10^{-2}z$, $z \sim \mathcal{N}(0, I)$, so seed-to-seed spread measures sensitivity to the starting point, not model or parameter uncertainty.

Table 1. Default medium and simulation parameters. Medium values are illustrative acrylamide/PVA-like choices inside cited ranges, not a refit of one published medium.

Parameter	Default	Source
D_0	$0.1 \mu\text{m}^2/\text{s}$	illustrative; AA/PVA literature value 0.01 [8]
σ	$0.08 \mu\text{m}$	illustrative; literature range 0.01–0.04 μm [10]
κ	2.0	calibration choice (cures the medium to $N \approx 1$ at unit dose)
γ	1.0	within the literature range 0.5–1.0 [9]
Δn_{max}	3.5×10^{-3}	within the AA/PVA range $1.5\text{--}5 \times 10^{-3}$
$k_{\text{bleach}}, \alpha_D$	0.2, 1.0	order of magnitude, not independently sourced
β	1.5	shape constant of Eq. (5) (this work)
s	0.5%	placeholder of typical AA/PVA magnitude
T, n_0, λ, ϕ	30 μm , 1.5, 405 nm, 0°	operating point; slant swept in Section 4.6
Grid	$n_x = 1024$, $\text{dx} = 0.05 \mu\text{m}$	51.2 μm periodic window; $n_z = 128$

3. Methods compared and benchmark

All methods share the exposure parameterization, the projection onto $\mathcal{E}$ and scoring on the full twin. **MIL** optimizes Eq. (6) with the twin (NPDD and beam propagation) inside the loss. **Media-blind SGD (BSGD)** runs the same optimizer under the linear recording assumption $\Delta n = \Delta n_{\text{max}}(E - \bar{E})$ without shrinkage and scores its exposure on the twin. **Gerchberg–Saxton (GS)** is alternating-projection phase retrieval [24, 25], its phase converted to an exposure by the same linear assumption. **Linear pre-compensation (LPC)** boosts each spatial-frequency

component of the target by $1/H(K)$, with $H(K)$ the heuristic small-signal transfer function of Supplement 1, Section S1, and then projects. **Gamma pre-compensation (GPC)** inverts a fitted pointwise saturating map. **TV-regularized SGD (RSGD)** is BSGD with a total-variation penalty whose weight was tuned on the twin per budget; the tuned weight is zero at all three budgets, so RSGD reduces to BSGD. The **saturation-only surrogate (SAT)** optimizes against a pointwise saturating model with all transport removed, calibrated to the twin by one fitted parameter (NRMSE 0.103). The **constrained oracle (ORC)** optimizes against an ideal linear medium under the same exposure constraints; it is a gradient-optimized reference, not a proven bound.

The benchmark target is a binary bar pattern at spatial frequency K , with periods rounded to an even number of pixels so that each nominal K is rendered exactly. The main grid covers 10 spatial frequencies, $K \in \{1.96, 2.62, 3.49, 3.93, 4.19, 4.83, 5.71, 6.98, 8.98, 15.71\}$ rad/ μm , at budgets $B_c \in \{2, 4, 8\}$, i.e. 30 cells, each with every method complete. Ablation and robustness studies use $K = 1.31, 3.93$ and 5.24 rad/ μm at $B_c = 2$. Iterative methods run with three seeds and deterministic ones (GS, LPC, GPC) once. The paired gain of a cell is the mean over seeds of $\text{PSNR}_{\text{si}}(\text{MIL}) - \text{PSNR}_{\text{si}}(\text{BSGD})$ at equal seed; intervals are two-sided 95% Student- t intervals over seeds unless a bootstrap [26] is stated. All simulations ran on one laptop (NVIDIA RTX 3050 GPU; the factorial, compute-matched and parameter-sensitivity jobs on its CPU).

4. Results

4.1. Paired gain across spatial frequency and budget

The paired gain is positive in 29 of 30 cells and its 95% interval excludes zero in 27 (Fig. 1(a)). The largest cell means are 1.42, 2.13 and 2.52 dB at $B_c = 2, 4$ and 8, all at $K \leq 2.62$ rad/ μm ; the grid means are 0.40/0.70/0.88 dB. At the highest tested $K = 15.71$ rad/ μm the gain is 0.01, 0.04 and 0.05 dB. One cell is negative: $K = 4.19$ rad/ μm at $B_c = 2$, -0.03 dB (95% interval [-0.05, -0.005]). The seed intervals are narrow because the seed changes only the initialization (Section 4.7 treats model error). Diffraction efficiency shows the same budget ordering (mean paired gain 1.80/3.00/3.94 percentage points) but 15 of 90 per-seed pairs are negative (minimum -0.251 points).

A heuristic linearization of the recording equations (Supplement 1, Section S1) predicts a spatial frequency $K_c(B_c)$ beyond which the pre-boost needed to undo the medium's roll-off exceeds the realized contrast, and hence a sign change of the gain: $K_c = 4.22, 6.96$ and 9.33 rad/ μm (dashed lines in Fig. 1(a)). At $B_c = 2$ the gain crosses zero at $K^* = 4.18$ rad/ μm (linear interpolation), close to K_c , but is positive again at every larger tested K , and at $B_c = 4$ and 8 it never crosses. The data therefore show a decay of the gain with K , not monotone at this sampling, rather than the persistent sign change the heuristic predicts. The constrained oracle's own advantage over BSGD falls with K as MIL's does, which we read, without a separate test, as a shrinking ceiling for any exposure-constrained method rather than a failure specific to MIL.

Figure 2 shows single-seed reconstructions at the three K of the ablation studies; Supplement 1, Fig. S1, shows the corresponding exposures and index profiles. At the lowest K MIL recovers most of the bar contrast that BSGD loses; at the highest, both reconstructions are similarly shallow.

4.2. Failure rate against an oracle reference

Each exposure of a write-once medium consumes material, so we also count how often a method falls far short of what is achievable: an exposure fails if its PSNR is more than a threshold below the constrained oracle's in the same cell (Table 2; the 2 dB threshold was fixed before the results were computed). At 2 dB, 37% of BSGD cells and 0% of MIL cells fail, and at 3 dB 17% and 0%; at 1 dB the rates are similar (60% and 57%). MIL removes the large shortfalls, not the small

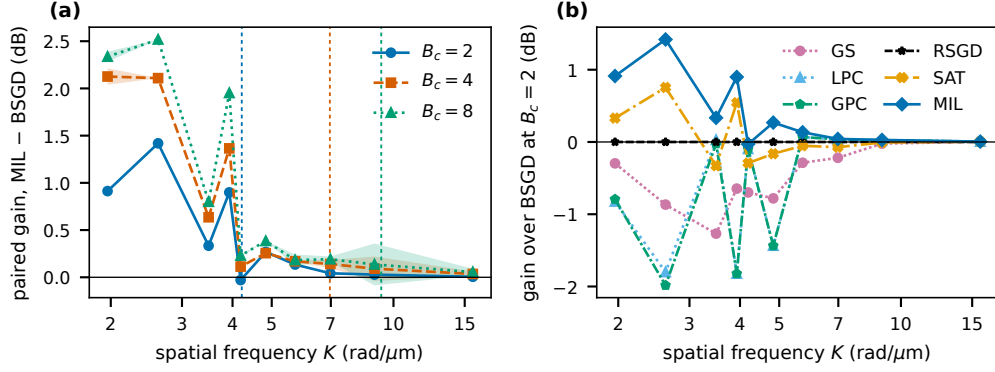


Fig. 1. (a) Paired gain (MIL - BSGD) versus spatial frequency for three contrast budgets; shading, 95% t -intervals over three seeds; dashed lines, heuristic $K_c(B_c)$. (b) Gain of every non-oracle method over BSGD at $B_c = 2$ (seed means); RSGD coincides with BSGD.

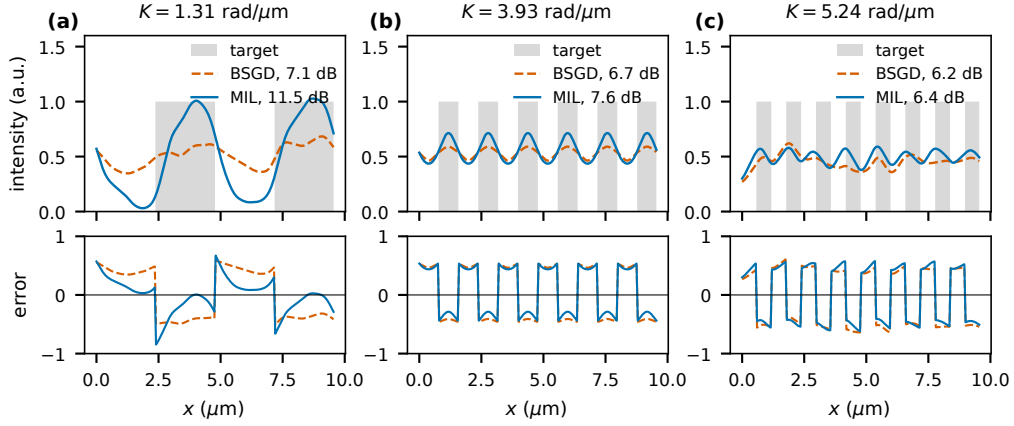


Fig. 2. Target, BSGD and MIL reconstructions (top) and errors (bottom) at $B_c = 2$, seed 0, over a $9.6 \mu\text{m}$ section of the window; legends give PSNR_{st} . Single-seed illustrations; Fig. 1 carries the statistics.

ones. The rates are relative to a numerical reference over 30 cells, not to a proven optimum or over physical trials.

Table 2. Failure rate: percentage of the 30 grid cells in which a method's seed-mean PSNR is more than the column threshold below the constrained oracle's.

Threshold	GS	LPC	GPC	BSGD	RSGD	SAT	MIL
1 dB	60	60	60	60	60	60	57
2 dB	53	47	47	37	37	33	0
3 dB	33	33	30	17	17	0	0

4.3. Other baselines and the saturation-only surrogate

At $B_c = 2$ (Fig. 1(b)), GS (-1.27 to 0.00 dB) never improves on BSGD and LPC (-1.82 to 0.07 dB) improves on it by at most 0.07 dB; GPC (-1.98 to 0.07 dB) differs from LPC by at most 0.18 dB, consistent with both hitting the same contrast clip. LPC is the closed-form use of the heuristic transfer function, so its failure is further evidence that the small-signal picture does not describe this regime. The saturation-only surrogate tests whether transport needs to be modelled at all: averaged over K , its gain is 18%, 28% and 42% of MIL's at $B_c = 2, 4$ and 8, and by cell it recovers 36–44% of MIL's gain at $K = 1.96$ rad/ μm and 53–66% at $K = 2.62$ rad/ μm but scores below BSGD in 12 of the 29 cells where the ratio is defined. A pointwise surrogate is therefore useful only at the lowest spatial frequencies; elsewhere the full recording model is needed for any gain.

4.4. Compute-matched control

MIL's per-iteration cost is about $21.7\times$ that of BSGD (one wall-clock measurement). With $21.7\times$ more BSGD iterations at the three ablation K (Supplement 1, Section S6) the paired gain is 3.19, 0.81 and 0.12 dB at $K = 1.31, 3.93$ and 5.24 rad/ μm , against 4.41, 0.90 and 0.11 dB at equal iteration count: the extra iterations narrow the gap at the two lower K but it stays positive at all three. Wall-clock time is a noisy proxy for compute, so this tests a much larger baseline budget rather than an equal-FLOP comparison.

4.5. Which recording nonlinearity produces the gain

Removing one mechanism at a time (Table 3, top) leaves a positive gain at every K , and which removal matters depends on K : linearizing the index map raises the gain at $K = 1.31$ and 5.24 but lowers it at 3.93 rad/ μm . Because three mechanisms saturate the dose response, removing one leaves the others, so we also ran all 2^3 on/off combinations of monomer depletion (removed by holding $u = 1$), dye bleaching ($k_{\text{bleach}} = 0$) and the tanh map (Supplement 1, Section S2). Removing monomer depletion at fixed κ raises the polymer cured by unit dose from $N \approx 1$ to $N \approx 8.7$, so these conditions are run with κ rescaled to restore $N(E=1) = 1$. A final control removes all three and sets $\kappa = 1/(1.5 t_{\text{total}})$, which makes the recording BSGD's own linear model up to non-local blur and shrinkage. The factorial ran on CPU and reproduces the GPU full-model values to within 0.001 dB.

Table 3. Paired gain (dB) under mechanism ablations, $B_c = 2$, mean of three seeds. Top: one mechanism removed. Bottom: saturating mechanisms isolated at a matched operating point, $N(E=1) = 1$, and the slope-matched linear control; blur, diffusion and shrinkage are kept. Intervals: Supplement 1, Figs. S2–S3.

Condition	$K = 1.31$	$K = 3.93$	$K = 5.24$
Full model	4.41	0.90	0.11
No non-locality ($\sigma = 0$)	4.47	0.82	0.05
No diffusion ($D_0 = 0$)	4.93	1.14	0.10
No dye bleaching ($k_{\text{bleach}} = 0$)	4.36	0.84	0.11
No index saturation (tanh linearized)	5.39	0.61	0.66
Only monomer depletion	5.75	1.73	1.26
Only dye bleaching	1.98	0.42	-0.02
Only tanh map	2.80	0.53	0.12
Dye bleaching and tanh map	3.64	0.72	0.08
Linear, slope-matched	-0.00	0.00	0.10

Three observations follow. First, when the recording is the media-blind model up to blur and shrinkage, MIL gains essentially nothing at the two lower K , so the gain there requires a nonlinear dose response; at $K = 5.24 \text{ rad}/\mu\text{m}$ the linear control's gain (0.10 dB) is comparable to the full model's (0.11 dB), so the small gain there may come from the blur and shrinkage that BSGD ignores. Second, each saturating mechanism alone produces a gain at the two lower K , so no single mechanism is necessary. Third, monomer depletion alone gives a larger gain than the full model at every K , so the mechanisms' effects do not add. In this model, then, the low- K gain comes from designing through whichever dose-response nonlinearity is present. The matching criterion $N(E=1) = 1$ is our choice; another normalization could reorder the single mechanisms, though not the null result of the linear control. Unmatched conditions are discussed in Supplement 1, Section S2.

4.6. Dependence on grating slant

With everything else fixed, the paired gain depends strongly and non-monotonically on the fringe slant ϕ (Fig. 3(a); Supplement 1, Section S3). At $\phi = 0^\circ, 5^\circ, 10^\circ$ and 20° it is 0.913, 2.077, 0.089 and 0.024 dB at $K = 1.96 \text{ rad}/\mu\text{m}$ and 0.899, 2.045, 0.185 and 0.011 dB at $K = 3.93 \text{ rad}/\mu\text{m}$: small slants increase the gain and larger ones remove most of it. The scalar readout disagrees with RCWA for slanted gratings (Section 6), so these values are model-dependent; this is why the headline results are restricted to unslanted gratings and should not be read as predictions for slanted holograms.

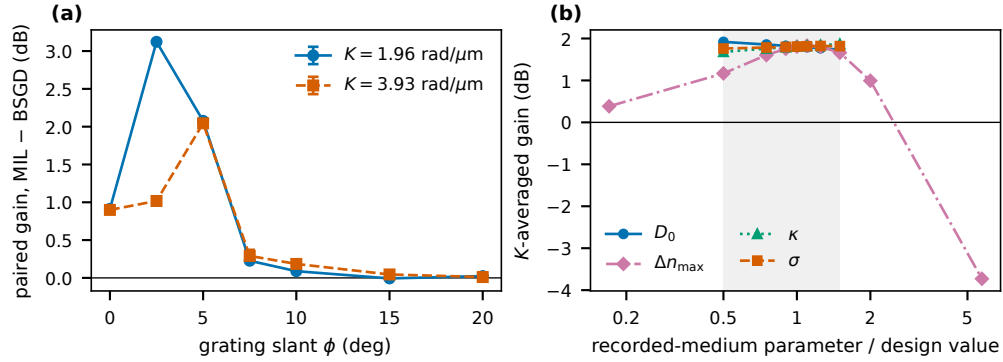


Fig. 3. (a) Paired gain versus grating slant, $B_c = 2$, 95% t -intervals over three seeds. (b) Gain when both exposures are designed with nominal parameters and recorded on a twin with one parameter scaled, averaged over the three ablation K ($B_c = 2$; shading, 95% intervals; grey band, $\pm 50\%$).

4.7. Robustness to twin miscalibration

One parameter. Here both exposures are designed once with the nominal parameters and recorded, without re-optimization, on a twin with D_0 , σ , κ or $\Delta n_{\max}$ perturbed by ± 10 , ± 25 or $\pm 50\%$ (Fig. 3(b); Supplement 1, Section S4). BSGD's design uses no recording parameter (only $\Delta n_{\max}$, as the scale of its linear model), so the test measures how much MIL's advantage depends on a correct twin. The K -averaged gain ranges from 1.17 to 1.92 dB (nominal 1.81 dB). We swept $\Delta n_{\max}$ further because two fits of the twin to Bayfol HX data disagree on it by $5.7\times$ (Section 5): the gain is 0.38 dB at -83% , 1.66 dB at $+50\%$, 0.99 dB at $+100\%$ and -3.73 dB at $+473\%$. A twin that underestimates the medium's index budget several-fold can therefore make MIL worse than the media-blind baseline; where between $+100\%$ and $+473\%$ the sign changes was not resolved.

Joint and structural errors. In 30 Monte Carlo draws with all four parameters perturbed independently and uniformly within $\pm 25\%$ (same designed exposures), the K -averaged gain ranges from 1.61 to 1.94 dB (mean 1.76 dB, bootstrap 95% interval [1.73, 1.79]); 0 of 30 are negative. Beer–Lambert attenuation through depth (16 recorded slices) gives 1.81, 1.51 and 1.45 dB at optical density 0, 0.1 and 0.3. Re-optimizing both arms on media with D_0 , σ or κ changed by up to $\pm 50\%$ ($K = 5.24 \text{ rad}/\mu\text{m}$) gives 0.08 to 0.15 dB (nominal 0.11 dB).

Targets and noise. At $K = 3.93 \text{ rad}/\mu\text{m}$ and $B_c = 2$ the gain is 0.90 for bars, 3.22 for sparse random spots and 0.17 dB for a random binary target, and Gaussian detector noise at 5% of the peak changes the bar gain from 0.90 to 0.88 dB (Supplement 1, Section S8).

5. Twin validation against published data

We fit the twin to the first-harmonic index modulation Δn_1 versus dose in Fig. 3 of Bruder et al. [5] (Bayfol HX, $\Lambda = 700 \text{ nm}$, $K = 8.98 \text{ rad}/\mu\text{m}$): the source’s kinetic-model curve ($16.7 \text{ mW}/\text{cm}^2$) and its measurement ($19.2 \text{ mW}/\text{cm}^2$), read visually rather than with a calibrated digitizer (about $\pm 10\%$ in dose, $\pm 3\text{--}5\%$ in Δn_1). Unfitted parameters take Bayfol HX values from the released configuration; each fit is the best of 8–10 random starts, and NRMSE is the root-mean-square residual over the range of the fitted data.

Fits and held-out prediction. With κ , $\Delta n_{\max}$ and k_{bleach} free, the twin reproduces the rise-then-decline shape of both series (Fig. 4; in-sample NRMSE 0.02 for the model curve, 0.17 for the measurement). Both are plotted against dose and the twin is dose-reciprocal at $\gamma = 1$, so a fit to one series predicts the other without refitting, with NRMSE 0.40 (model to measurement) and 0.31 (the reverse). The fitted k_{bleach} are 0.0667 and 0.0215 (default 0.2) and the fitted $\Delta n_{\max}$ 0.0418 and 0.0321. With k_{bleach} held at 0.2 the in-sample NRMSE is 0.17 and 0.28, but the measured-series optimum is degenerate (its κ saturates the medium before the first measured dose, giving an almost constant Δn_1 ; Supplement 1, Fig. S5), the held-out NRMSEs are 0.48 and 1.09, and the fitted $\Delta n_{\max}$ disagree by $5.7\times$, which sets the range of the $\Delta n_{\max}$ sweep in Section 4.7. Fig. S5 also shows a fit to a PQ/PMMA $\Delta n_1(t)$ model curve [6] outside the tested K range (NRMSE 0.10).

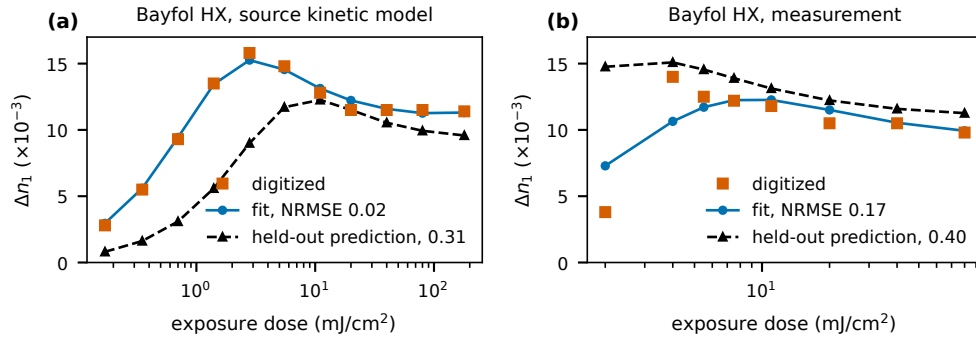


Fig. 4. Twin fitted to Δn_1 digitized from Bruder et al. [5] (Bayfol HX): (a) the source’s kinetic-model curve, (b) its measurement. Solid, fit to the series shown (κ , $\Delta n_{\max}$, k_{bleach} free); dashed, parameter-free prediction from the fit to the other series. Legends give NRMSE.

What this establishes. The model family can reproduce, and predict across two views of the same condition, the dose response of one commercial photopolymer with effective parameters. It does not pin the default medium of Table 1 to a real material or test transfer across K or media: the two series share a material, a grating and nearly the same intensity, and one is a model output. D_0 and σ are identifiable only from multi- K growth-curve families, which we did not have. The

headline claims therefore rest on the paired design and the miscalibration tests, not on absolute agreement with a real medium.

6. Discussion and limitations

Scope. All results are simulation outputs of one model family for one-dimensional transverse recording of binary bar targets in unslanted gratings, with one default medium and three seeds per cell. Whether the gains transfer to a physical photopolymer is untested; the decisive experiment, recording optimized and media-blind exposures in a characterized medium and comparing the measured reconstructions, is future work.

Readout validity. Against RCWA [20] (computed with TORCWA [27]) over a 90-case grid of K , Δn , polarization and slant, Kogelnik theory deviates by at most 0.29 in diffraction efficiency for unslanted geometries but by up to 0.98 at a 20° slant (Supplement 1, Section S5). The check concerns the closed-form reference, but the beam propagation in the design loop shares its paraxial scalar approximation, so validated claims are restricted to unslanted gratings, a restriction that matters (Section 4.6).

Model and statistics. The recorded profile is uniform through depth except in the Beer–Lambert check; diffusion uses a spatially averaged coefficient; the index map and its shape constant are modelling choices; shrinkage is a placeholder; the medium parameters are illustrative; and the factorial covers interactions among the saturating mechanisms only. Three seeds per cell measure initialization sensitivity, which is small here; model and parameter error are probed only by the miscalibration studies over the stated ranges. The 10-point grid does not resolve features between grid points, such as the zero crossing at $B_c = 2$.

Implications for practice. If the twin’s qualitative response is right, exposure-domain optimization through a recording model is most useful at low spatial frequency and in unslanted geometry, and its main benefit is removing large shortfalls (Table 2) rather than raising average quality. Calibrating the twin for a new medium needs κ and $\Delta n_{\max}$ from a dose-response curve measured under the same condition, D_0 and σ from several spatial frequencies, and a fitted dye-bleaching rate (Supplement 1, Section S7); a several-fold underestimate of $\Delta n_{\max}$ is the error most likely to reverse the benefit.

7. Conclusion

In simulation, designing the exposure of a write-once volume photopolymer through a differentiable NPDD and beam-propagation model improves on media-blind design in 29 of 30 unslanted grating cells, by up to 2.52 dB at low spatial frequency, and removes the exposures that fall more than 2 dB short of an oracle reference. The gain requires a nonlinear dose response but no single saturating mechanism, decays toward zero at high spatial frequency and under fringe slant, survives joint parameter errors of $\pm 25\%$, and can reverse if the twin underestimates the index budget several-fold. Testing these model predictions in a recorded medium is the next step.

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Disclosures. The author declares no conflicts of interest.

Data availability. Code, experiment manifests, raw per-job result files and the scripts that generate every number and figure in this paper are available on GitHub (<https://github.com/ultramagnus23/HoloForge>) and archived in Ref. [28].

Supplemental document. See Supplement 1 for supporting content.

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